work," Nair reports. The students downloaded about 600 GB of videos and converted them into OGV files. Nair explains, "Most videos online use the MP4 format, which is proprietary and comes with certain restrictions, but OGV is an open source format under a free software licence. This format is supported by the Free Software Foundation. With OGV, we didn't have to use any proprietary software to encode all videos. We used an HTML5 standard, which had a *VIDEO* tag, which enabled supporting browsers to play the video natively in the browser without requiring any external plug-ins like Flash." Moreover, converting files to this format also compressed videos to about 40 per cent of their actual size. This made a lot of difference, as each course took about 5 GB of disk space.

When the students started their academic year, they were used to working on MySQL. Their professors, Dr Madhu Kumar S D and Gopakumar G, suggested that they start working on PostgreSQL. "When we asked why, our professors asked us to experiment and find out the answer," says Vipin Nair. The students soon realised that PostgreSQL was much richer in features than MySQL, and later decided that it would be apt for the video repository too.

The volume of data cannot be ascertained yet, as the project is only due for release next month. "But we are expecting students to visit our site often, especially before our exams, as that is when most of us get to work," jokes Nair. In terms of concurrency, there are over 6,000 students in the college now, and the group expects a maximum of 2,000 to 3,000 to access the videos at any given time. "It will peak during exams, but our bandwidth is currently limited. If we get dedicated lines for the repository, we can serve a higher number. Right now, we are operating from a room with computer centres, and share bandwidth lines with the college labs. Scalability is not an issue for us, because at any point of time, we are not expecting over 10,000 videos. In a decade, we may increase the number of videos to a lakh, but even that is a small number for any database," Nair confidently states.

Where PostgreSQL scores

After taking the time to understand and configure PostgreSQL, the students are satisfied with the database. The other option they considered was MySQL, but it comes with a default MyISAM engine that does not offer integrity constraints. "For example, a video in the database will have tags for itself. If the tags are only for that video, I would like the loose tags to get deleted when I delete the video—but this is not possible with the MySQL default engine. PostgreSQL scores in this aspect, and ensures that there are no loose tags stored. Another feature that pushed us towards PostgreSQL is that it supports Object-Oriented Databases, while MySQL has no support for object-oriented database concepts. Our database design had table inheritance, and PostgreSQL supports such designs natively; although it has a few bugs, I still consider it a good option," says Nair. He also believes that PostgreSQL has the majority of enterprise-level features, compared to other databases. Apart from integrity, it provides other useful features like schemas, which is useful while handling mission-critical work. "This feature allows us to give different people limited access to the database and, hence,

Paathshaala is an an internal repository of tutorial videos for the students of NIT-C college

provides flexibility to keep our data secure," says a contented Nair.

The challenges

The students faced some problems regarding PostgreSQL in the implementation and installation stages, but they solved them easily by reading up on the matter. Nair explains further, "We usually update from our command line, and in MySQL, it is pretty easy to do so. However, in this case, we were not familiar with the procedure, and there were too many extra steps. This was solved as soon as we read up on the topic, and we worked on it quickly. Another problem we are facing regarding the project is that our site may slow down when too many people access it at the same time. We are talking to our network admins about this problem, and we hope that they will give us a dedicated line in such a situation."

Support system

At present, the group that forms Paathshaala has taken on the responsibility for providing support. "Once it is launched, we will bring our juniors into the group. We have documented every design aspect, policy decision, and all the necessary details of the project. We will pass it on to our juniors, who will handle the project when we pass out. We will also release the source code of this project as free software under GPLv3, which will help any student willing to contribute to the Paathshaala code base," explains Nair. Moreover, team members constantly interact with the online FOSS community, and their teachers have assisted them whenever they needed guidance.

The road ahead

In the future, NIT-C plans to make the video repository similar to YouTube, where students can upload content, rate it, and contribute their comments. They may even start producing their own videos and uploading them on the site. Who knows, students from the Harvards and Stanfords of the world may soon start referring to their lectures in the years to come! END

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